Stored Procedures

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```
m1(){
  -
  -
  -
  -
  -
}
```

m1()
m1()
m1()

Reusable

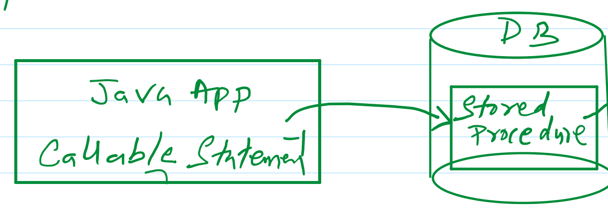
Database Programming

procedure addRecordsInEmpPaymentsActivities():
Begin
-
-
-
-
End.

→ add one record in employee table
→ add one record in Payments table
→ add one record in Activity table
:
:

best reusable components.

group of SQL queries.



ORACLE → PL/SQL

MySQL → Procedure language.

Microsoft SQL Server → TSQL (Transact SQL).

```
public int m1(int x, int y){
    int z = x+y;
    return z;
}
```

How many types of parameter stored procedure takes?

1. IN parameter (To provide input value)
2. OUT parameter (To collect output value).
3. INOUT parameter (To provide input values and to collect output values)

eg: $z := x + y$ ⇒ x & y are IN parameter
⇒ z is OUT parameter.

$x := x + 1$ ⇒ x is INOUT parameter.

What is Stored Procedure?

In our programming if any code repeatedly required, then we can define that code inside a method and we can call that method multiple times based on our requirement.

Hence method is the best reusable component in our programming.

Similarly in database programming, if any group of SQL statement

Hence method is the best reusable component in our programming.

Similarly in database programming, if any group of SQL statement is repeatedly required then we can define those SQL statements in a single group repeatedly based on our requirement.

This group of SQL statements that performs a particular task is nothing but Stored Procedure. Hence stored procedure is the best reusable component at database level.

Hence, stored Procedure is a group of SQL statements that performs a particular task.

These procedures stored in database permanently for future purpose and hence the name stored procedure.

Usually stored procedure are created by Database Administrator (DBA)

Every database has its own language to create Stored procedures.

Stored procedure in JDBC (Java database connectivity)

A stored procedure is a precompiled collection of SQL statements stored inside the database. It performs one or more database operations and can accept input parameters, return output parameters, and even return result sets.

Instead of sending SQL queries from a Java program every time, the Java application simply calls the stored procedure.

In JDBC, stored procedures are executed using the **CallableStatement** interface.

Why use stored procedures?
Consider an application where the salary of an employee needs to be updated.

Without stored procedure
The Java program sends the SQL query every time.

```
SQL:
UPDATE employee
SET salary = salary + 5000
WHERE emp_id = 101;
```

The SQL statement is parsed, compiled, optimized and executed each time.

With Stored Procedure
The SQL statement is stored permanently inside the database.

```
SQL:
   CALL increase_salary(101, 5000);
```

Only the procedure name and parameters are sent from Java, making execution faster.

Advantages of stored procedures

1. Faster execution because they are precompiled.
2. Reduces network traffic.
3. Improves security by hiding SQL queries.
4. Reusable by multiple applications.
5. Easier maintenance.
6. Reduces code duplication
7. Supports complex business logic.

How to create and execute stored procedure in Oracle?

```
create or replace procedure proc1(X IN number, Y IN number, Z OUT number) as
BEGIN
   Z:= X + Y
END
```

Notes:

SQL and PL, SQL are not case sensitive languages. We can use lower case and upper case also.
After writing stored procedure we have to compile. For this we require to use /(forward slash)

/ => compilation

While compiling if any error occur then we can check these errors by using the following command.

```
SQL> show errors;
```

Once we created stored procedure and compiled successfully, we have to register out parameters to hold result of stored procedure.

```
SQL> variable sum number; (declaring a variable)
```

We can execute the stored procedure with **execute** command as follows:

```
SQL> execute procedure(10, 100, sum);

SQL> print sum;
```

Practical implementation:

```
SQL> create or replace procedure p1(x IN number, y IN number, z OUT number) as
  2  Begin
  3     z:=x+y;
  4  End
  5  /

Warning: Procedure created with compilation errors.

SQL> show errors
Errors for PROCEDURE P1:

LINE/COL ERROR
-------- -----------------------------------------------------------------
4/3      PLS-00103: Encountered the symbol "end-of-file" when expecting
```

```
         one of the following:
         ; <an identifier> <a double-quoted delimited-identifier>
         current delete exists prior <a single-quoted SQL string>
         The symbol ";" was substituted for "end-of-file" to continue.

SQL> create or replace procedure p1(x IN number, y IN number, z OUT number) as
  2  Begin
  3     z:=x+y;
  4  End;
  5  /

Procedure created.

SQL> variable sum number;
SQL> execute p1(2.5, 7.9, sum);
BEGIN p1(2.5, 7.9, sum); END;

                   *
ERROR at line 1:
ORA-06550: line 1, column 20:
PLS-00201: identifier 'SUM' must be declared
ORA-06550: line 1, column 7:
PL/SQL: Statement ignored
Help: https://docs.oracle.com/error-help/db/ora-06550/


SQL> execute p1(2.5, 7.9, :sum);

PL/SQL procedure successfully completed.

SQL> print sum

       SUM
----------
      10.4

SQL> execute p1(2.5, 7, :sum);

PL/SQL procedure successfully completed.

SQL> print sum

       SUM
----------
       9.5
```